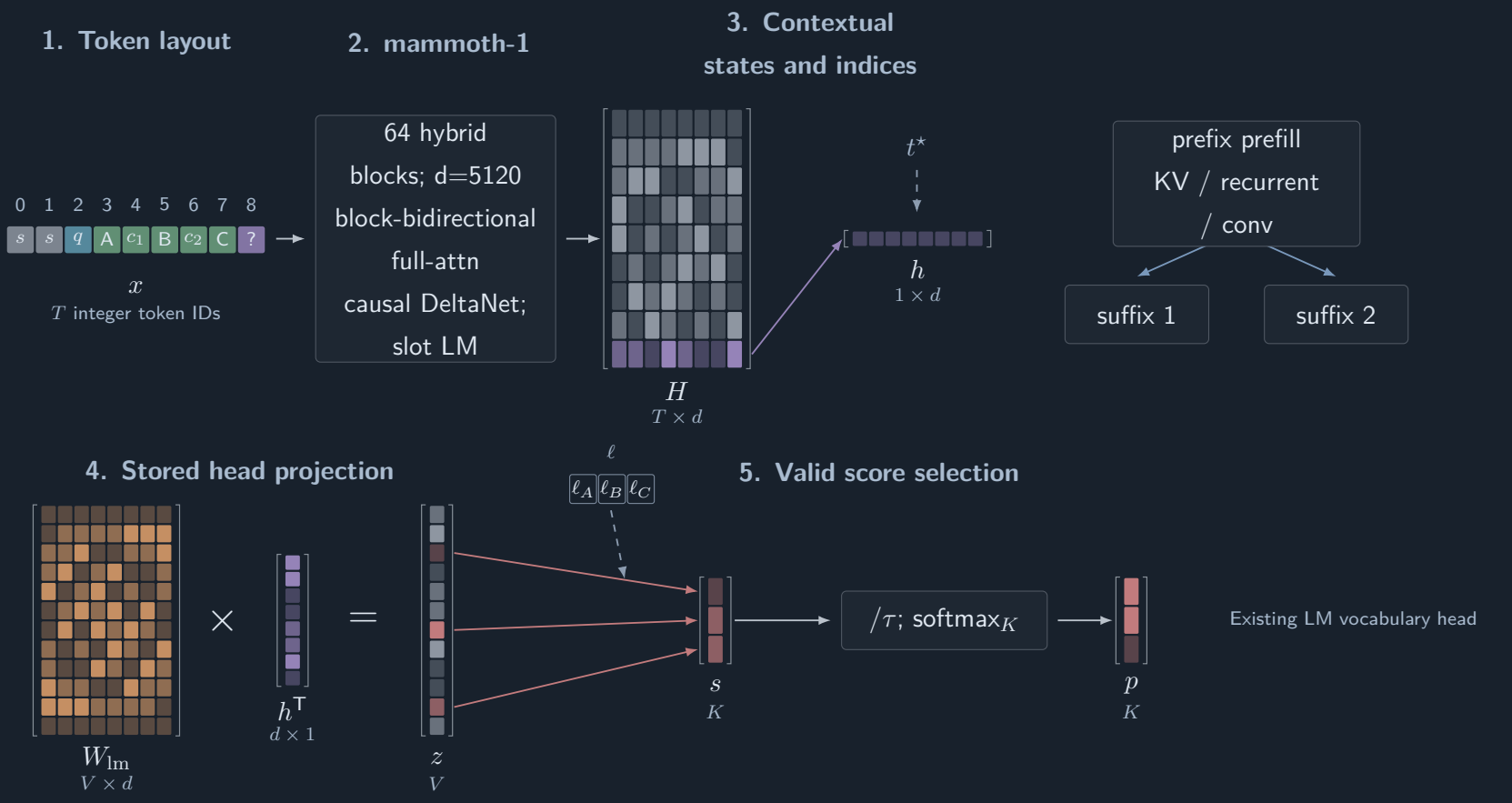
$$h = H_{t^\star,:}, \quad s_k = \langle W_{\ell_k,:}, h \rangle, \quad p = \mathrm{softmax}_K(s/\tau)$$



Axes $d = 5120$, $V = 248320$; Single-question illustration; batch omitted. Tile counts encode axes, not literal sizes.

Objects Gray: input/context; violet: hidden states; orange: parameter matrices; red: logits/probabilities. Positions, code IDs and owner maps are integers, not scores. Tiles show roles, not measured magnitudes.

Mechanism One shared prefix, separate suffix rows and cache forks. The optional pointer head_dim=256 is not a 5120-to-256 MLP in the released slot_lm model.

Existing LM head, then slot-ID gather; 256 declared slots. Source loss is candidate CE plus sum-slot Brier, batch mean; released-run settings are author reports.